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D.T.A. Symons

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Early Permian pole: Evidence from the Pictou red beds, Prince Edward Island, Canada

D.T.A. Symons Department of Geology, University of Windsor, Windsor, Ontario, Canada N9B 3P4

ABSTRACT

The mid-Permian red beds of the upper Pictou Group retain a primary remanence in hematite that has been isolated by thermal step demagnetization above 600 °C. Conglomerate, contact, and reversals tests show that the remanence is primary. The pole position of lat 42.1°N, long 125.5°E (dp = 1.8°, dm = 3.6°) falls on current apparent polar wander paths at older ages than expected. There is also a significant difference between Early Permian poles from the northeastern United States–Canada and the southwestern United States. The results suggest that postdepositional Permian-Triassic remagnetization may have affected some units in the southwestern United States or that there has been about 9° of relative tectonic rotation between the two regions caused by the Laramide orogeny.

INTRODUCTION

The Permian apparent polar wander (APW) path for North America is based on ⩽10 paleomagnetic poles from flat red beds in the southwestern United States (Irving and Irving, 1982; Gordon et al., 1984). The reliability of some of these poles is limited by uncertainties in age or tectonics, by inadequate sampling or demagnetization, and by the lack of stability tests. For example, any of these red beds could retain a much younger remanence (Larson et al., 1982) like remagnetized beds elsewhere (e.g., Miller and Kent, 1986). A true reference path is important for plate reconstruction and terrane displacement studies, but also because many poles from older Paleozoic units are being assigned to the Kiaman reversed superchron (Pennsylvanian-Permian) on the assumption that they have been remagnetized to give a Kiaman pole (Irving and Strong, 1985; Stearns and Van der Voo, 1987). Although this assumption can be challenged (Symons and Lewchuk, 1989), this study was done to improve the precision of the Kiaman pole.

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Except for one sill outcrop, Prince Edward Island is entirely underlain by flat-lying (1°–4° dips) red beds of the Pictou Group (van de Poll, 1983). The group is divided into megacyclic sequences I to IV, each of which is 300–400 m thick. Each megacycle contains a basal conglomerate, many sandstone-siltstone cycles up to 20 m thick, and a fine siltstone capping. Paleontologic evidence defines a Stephanian (Late Pennsylvanian, 286–306 Ma) to Artinskian (mid-Early Permian, 263–268 Ma) age. The gentle dips reflect undated uplift over a basement horst. At Malpeque Bay (Fig. 1) the red beds are intruded by a 3-m-thick alkalic lamprophyre sill. Its convoluted baked contacts indicate that the red beds were unconsolidated when intruded (Larochelle, 1967). Greenough et al. (1988) obtained three Late Permian K-Ar whole-rock ages of 247 ±1 Ma for the sill and showed that younger ages by Wanless et al. (1973) resulted from argon loss. Previous paleomagnetic studies of the Pictou Group by Black (1964) used alternating field (AF) demagnetization on a small collection, and studies by Roy (1963, 1966) used thermal demagnetization to only 450 °C.

EXPERIMENTAL METHODS

Four oriented blocks were collected from each of the 18 sites in the Early Permian red beds of megacycle IV and from two sites in the "green" beds of small reduction zones (Fig. 1, Table 1). A further 12 blocks of weathered red beds were collected around a small fold with dips up to 12° for a fold test because fresh samples on the sea cliffs were inaccessible, but these gave poor results. Four 2.5 cm specimens were drilled from each block, sometimes by using a coating of sodium metasilicate to prevent disintegration (Symons et al., 1989). Also, 13 oriented walnut-sized clasts were collected for a conglomerate test, and two 0.9 cm specimens were drilled from each.

An automated CTF cryogenic magnometer was used to measure the natural remanent mag-

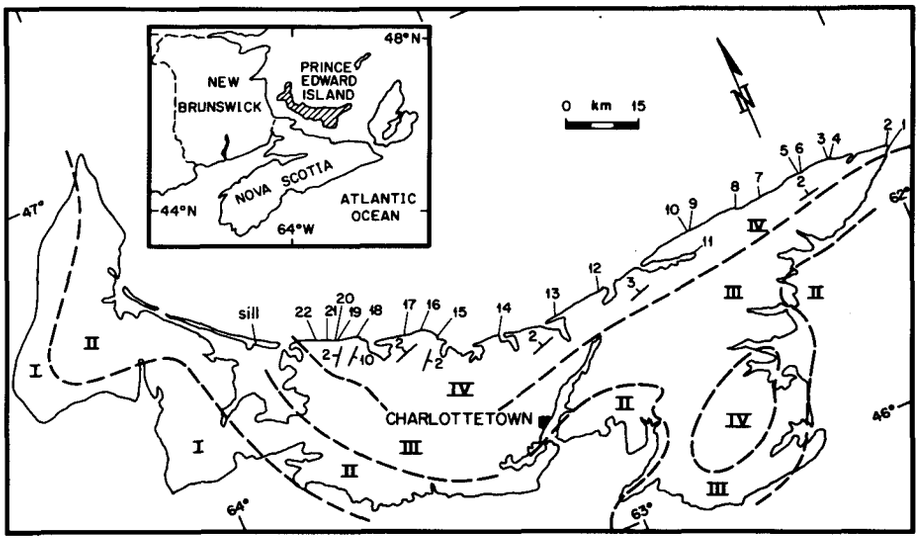


Figure 1. Geology of Prince Edward Island with sampling sites; I, II, III, and IV refer to megacycles in Pictou Group red beds (van de Poll, 1983).

TABLE 1. SITE MEAN EMANENCE DIRECTIONS AFTER TILT CORRECTION

Site no., Rock unit*	No. of blocks, No. of specimens	Mean Direction† Decl. (°)	Incl. (°)	α_{95} (°)	k
1, r	4, 13	181	-1	18	26
2, r	4, 15	173	-1	12	64
3, r	4, 16	173	1	21	20
4, r	4, 13	177	11	10	79
5, c	13, 26	25	70	68	1
6, r	4, 13	173	-13	9	109
7, r	4, 4	174	9	12	63
8, r	4, 15	172	3	14	46
9, r	4, 16	178	0	15	40
10, g	4, 9	174	18	14	43
11, r	3, 10	163	11	20	41
12, w	3, 8	169	9	33	15
13, r	4, 13	177	-3	12	60
14, r	4, 15	185	10	15	37
15, r	4, 11	173	-1	11	76
16, r	4, 15	167	9	15	40
17, w	3, 5	162	4	31	17
18, w	9, 13	168	-6	23	11
19, w	2, 7	176	-1	-	9
20, g	3, 4	184	11	25	26
21, r	4, 13	169	-2	20	23
22, r	4, 13	178	1	19	25

*Rock types: r = red beds; w = weathered red beds; g = bleached "green" beds; c = conglomerate clasts.
†Defined by declination, inclination, radius of 95% confidence (α_{95}) and precision parameter (k) (Fisher, 1953).

netization (NRM). The NRM intensities of the red beds are mostly in the low 10^{-3} A/m range; the green beds and clasts are in the low 10^{-4} A/m range. The NRM directions of the red beds are about horizontal and to the south-southeast in the Kiaman R component direction, whereas the green beds and clasts are scattered about the present Earth's magnetic field (PEMF) direction. A Sapphire Instruments SI3 unit was used to demagnetize one specimen from each red bed site to 150 mT, but most specimens were barely affected. After removing PEMF viscous remanent magnetization (VRM) components, the remanence of four green bed specimens swung to the R direction by 80 to 100 mT on AF step demagnetization (Fig. 2A), above which they required anhysteretic remanence (ARM). The clast specimens showed the linear decay of stable remanence vectors on AF demagnetization (Fig. 2B).

Two specimens from each red-bed site were thermally demagnetized in eight steps to 680 °C by using a Schonstedt TSD-1 unit. A few specimens showed isolation of the R direction with unlocking temperatures close to the 675 °C Néel temperature of hematite (Fig. 3A). Most showed the removal of some VRM before isolating the R direction above about 640 °C (Fig. 3B). Some, particularly the weathered red beds, had larger VRM components with lower blocking temperatures (Fig. 3C). All remaining specimens of red beds were demagnetized at 625, 655, and 670 °C, of green beds at 20, 50, 100, and 160 mT, and clasts at 20, 40, 60, and 80 mT.

The Sapphire Instruments SI3 was used to test isothermal remanence (IRM) of six typical specimens. The IRM acquisition curves for the red beds are typical of both fine- and coarse-grained hematite and for the green beds of fine-grained magnetite (Fig. 4A). The IRM decay curves are all close to the single-domain limit for both hematite and magnetite. Specimen directions were calculated by averaging two or more step directions that defined an end point. The specimen means were averaged to block means and these in turn to site mean directions (Fig. 5, B and C, Table 1).

REMANENCE STABILITY TESTS

A conglomerate test was run on 13 clasts of quartzite, rhyolite, and other rock types (Graham, 1949). Two specimen directions were averaged to get each clast direction (Fig. 5A). The resultant vector of the 13 clast directions of 4.03 is markedly less than the 1% significant point value of 6.84 (Watson, 1956). Thus, the test states with >99% confidence that the clasts have not been remagnetized since deposition, and therefore the Pictou red beds have a primary remanence.

Larochelle (1967) found after AF cleaning 24 specimens (12 blocks) that the Malpeque Bay sill and baked zone gives a mean direction of 182.1°, −16.4° (α_{95} = 5.2°), which differs significantly from the red beds (Fig. 6). Correcting for the 15° tilt of the sill, which may or may not be primary, moves the mean by only 2°. The uncorrected pole for the sill of 51.7°N, 112.8°E (dp = 3°, dm = 5°) agrees with its Late Permian age of 247 ±1 Ma. Thus, this is a positive contact test (Everitt and Clegg, 1962) that indicates that the remanence of the red beds predates 247 ±1 Ma.

Normal remanence directions were found in nine specimens from three blocks at site 2, 15, and 16. Their mean direction is antiparallel to the mean for the red beds (Table 2), and after reversal they are indistinguishable at the 95% confidence level (McFadden and Lowes, 1981),

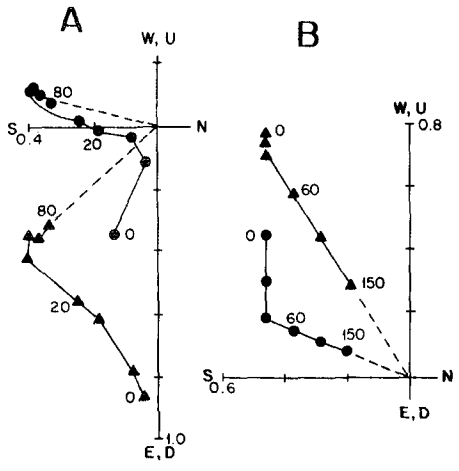


Figure 2. Orthogonal vector AF demagnetization diagrams for example specimens from sites 9 (A) and 24 (B). Circles and triangles denote horizontal and vertical projections, respectively. Axial intensities are ratio of NRM intensity. Step intensities are in milliTesla.

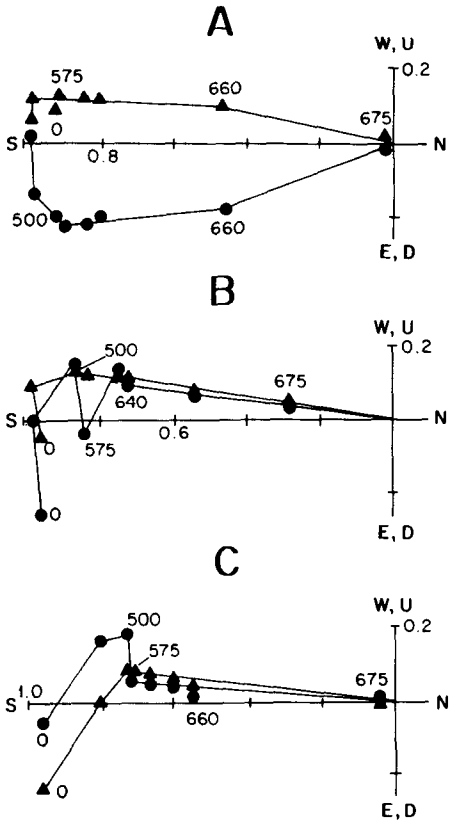


Figure 3. Orthogonal vector thermal demagnetization diagrams for example specimens from sites 15 (A), 8 (B), and 15 (C). Conventions as in Figure 2 except step intensities are in degrees Celsius.

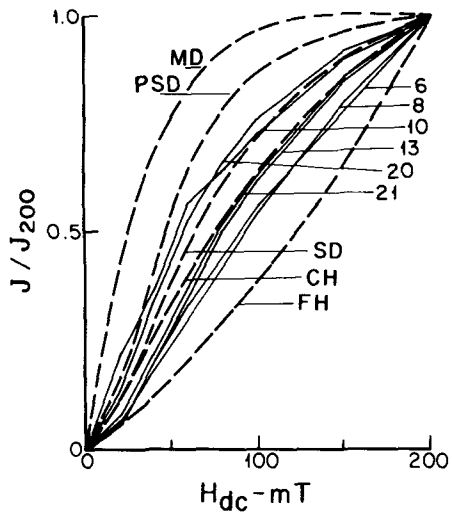


Figure 4. Isothermal remanent magnetization (IRM) data showing acquisition of IRM up to 200 mT. Type curves are for multidomain (MD), pseudosingle-domain (PSD), and single-domain (SD) magnetite and for coarse-grained (CH) and fine-grained (FH) hematite.

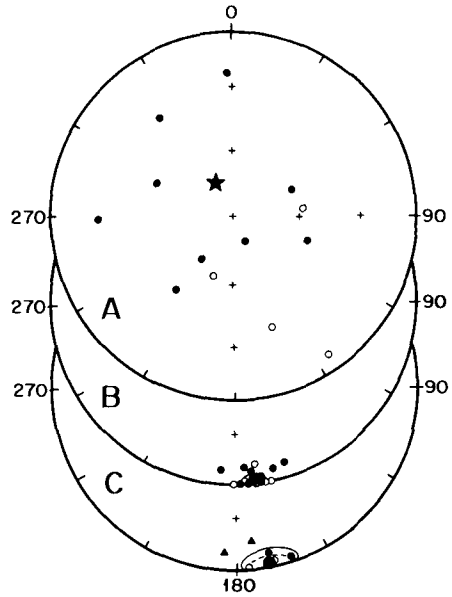


Figure 5. Equal-area stereograms showing (A) mean directions for 13 conglomerate clasts, (B) site means for red beds, and (C) site means for weathered red beds (circles) and green beds (triangles). Directions are after tilt correction. Solid and open symbols denote downward and upward vectors, respectively. Larger symbol is unit mean direction with its cone of 95% confidence. Star is present Earth's magnetic field.

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indicating the isolation of a primary remanence. At least two normal events are recorded, because site 2 is near the base of megacycle IV whereas sites 15 and 16 are near the top. Also, the fact that 96% of the specimens are reversed agrees with the global estimate of Irving and Pullaiah (1976) that the field was reversed for 95% of the Kiaman.

The green beds occur in small irregular reduction zones that follow fractures in the red beds. After removal of VRM, the nine blocks give a mean direction that is not significantly different from the red-bed mean when the differences in precision parameters (34 vs. 88) and calculation basis (block vs. site means) are included (McFadden and Lowes, 1981). The directions indicate that reduction took place in either Permian or Holocene time but not some intermediate time. If Permian, then the red-bed remanence is either primary or diagenetic; but if Holocene, then no limiting conclusion may be drawn.

Because tilts are small, dip correction does not significantly change the mean direction of the red beds or its precision (Table 1), and therefore the fold test is inconclusive (McFadden and Jones, 1981).

DISCUSSION

The pole of lat 42.1°N, long 126.5°E (1.8°, 3.6°) for the upper Pictou red beds (Table 2, line g) differs significantly from Roy's (1966) pole of 42°N, 133°E (3°, 6°) for the Pictou Group overall. However, it is identical to the pole of 42°N, 128°E (5°, 9°) calculated for his 10 accepted sites (A10 and A12–A20, 32 specimens) from megacycle IV, even though he used only 450 °C demagnetization, except that his error ellipse is significantly larger. This confirms the remanence stability of these red beds.

Irving and Irving (1982) accepted 10 poles (Table 3, poles 1–10) to define the Permian APW path (Fig. 6). To get the eight acceptable Permian poles for their APW path, Gordon et al. (1984) (1) accepted poles 7, 8, and 9; (2) rejected poles 3 and 4; (3) reworked the data for poles 1, 2, 5, and 10 into two poles; and (4) split the data for pole 6 into three poles. Also, their age assignments reflect minor revisions from following the Harland et al. (1982) time scale. Two obvious weaknesses of this data base are that all 10 poles come from red-bed sequences in the southwestern United States and that several poles were determined some 20 years ago with evident limitations by present standards (Table 3, Notes). Permian paleomagnetic data from elsewhere in North America are sparse. Pole 11 from 258 ±1 Ma Middletown pegmatites in Connecticut is uncertain because it lacks paleohorizontal control. Pole 12 from the Cap aux Meules red beds on the Iles de la Madeleine is thought by Tanczyk (1988) to average out both prolonged diagenetic alteration of the hematite and a concomitant Permian counterclockwise rotation. The virtual pole (14) from the Malpeque Bay sill may not average out secular variation. The pole from the Esayoo volcanics in the Canadian Arctic is omitted because of an inferred 36° ±8° rotation of the sampling area (Wynne et al., 1983). Also omitted are a few poles that may include some Permian strata but are dominantly in either older or younger strata such as the Dunkard Formation (Helsley, 1965) and the Vail section (Miller and Opdyke, 1985).

TABLE 2. UNIT MEAN REMANENCE DIRECTIONS AFTER TILT CORRECTION

	Group	Includes Table 1 Sites	Mean direction*						
			M	N	R	Decl. (°)	Incl. (°)	α_{95} (°)	k
a.	Red beds	r	S	15	14.8451	174.2	2.1	4.0	90
b.	Weathered red beds	w	S	4	3.9646	168.8	1.7	10.0	85
c.	Green beds	g	S	2		179	15.		
d.	Green beds	g	B	7	6.8222	178.2	15.1	10.5	34
e.	Normal beds	2, 14, 15	B	3	2.9415	355.1	-2.2	21.4	34
f.	Normal blocks	2, 14, 15	Sp	9	8.6043	354.5	0.3	11.7	20
g.	All red beds	r, w	S	19	18.7956	173.1	2.0	2.6	88
h.	All red beds	r, w	Su	19	18.7930	173.1	0.9	3.6	87
i.	All sites	r, w, g	S	21	20.7312	173.7	3.2	3.7	74
j.	All sites	r, w, g	Su	21	20.7317	173.6	2.1	3.7	74

*M = calculation method by site means (S), block means (B), specimens (Sp) or site means before tilt correction (Su); R = resultant vector; Decl., Incl., α_{95} and k as in Table 1.

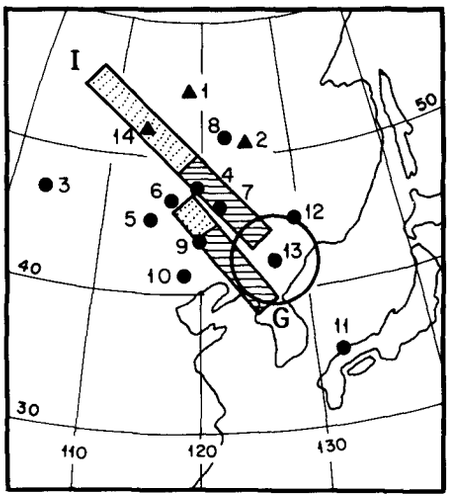


Figure 6. Permian pole positions as listed in Table 3. Circles and triangles denote Early and Late Permian poles, respectively. APW paths of Irving and Irving (1982; I) and of Gordon et al. (1984; G) are divided into Early (lines) and Late (dots) Permian segments. Upper Pictou Group pole (13) from this study is circumscribed by its oval of 95% confidence.

TABLE 3. PERMIAN POLE POSITIONS FOR NORTH AMERICA

No.	Rock Unit	Area*	Age†	Lat. (°N)	Long. (°E)	α_{95} (°)	Notes§
1.	Ochoan red beds	SW U.S.	L	55	119	8	a,f,h,k,n
2.	Guadaloupian red beds	SW U.S.	L	51	125	5	a,f,h
3.	Toroweap Formation	Ariz.	E	47	103	6	a,f
4.	Fountains and Lyckens Formations	Colo.	E	47	119	10	a,g,j,m,n
5.	Leonardian red beds	SW U.S.	E	45	115	13	a,f,h
6.	Cutler Formation	Colo.	E	46	117	5	a,h,m
7.	Ingleside Formation	Colo.	E	46	122	2	a,f,h
8.	upper Casper Formation	Wyo.	E	51	123	4	a,h
9.	Elephant Canyon Formation	Utah	E	44	120	5	a,f
10.	Wolfcampian red beds	SW U.S.	E	41	118	6	a,h,m
11.	Middletown pegmatites	Conn.	E	35	132	7	b,f
12.	Cap aux Meules Formation	Quebec	E	45	129	8	c,i
13.	upper Pictou Group	PEI	E	42	127	7	d,i,k,l
14.	Malpeque Bay sill	PEI	L	52	113	4	e,f,q

*PEI = Prince Edward Island.
†E = Early Permian; L = Late Permian.
§a = referenced in Irving and Irving (1982); b = deBoer et al. (1972); c = Tanczyhk (1988); d = this study; e = Larochelle (1967); f = pole from <50 specimen vectors; g = AF demagnetization of red beds only; h = thermal demagnetization of red beds to <600 °C; i = extensive thermal demagnetization at >600 °C; j = pre-Tertiary based on conact and fold tests; k = positive reversals test; l = positive contact and conglomerate tests; m = may include Pennsylvania strata; n = may include Triassic strata; o = Colorado Plateau, may be rotated 7° ± 4° and/or translated southward 4° ± 4° (Steiner, 1986; Bryan and Gordon, (1986); q = likely a virtual pole.

CONCLUSIONS

Field and other stability tests indicate that a stable primary Artinskian or mid-Early Permian remanence has been identified in the upper Pictou red beds. The resulting pole (13) is probably the most reliable Early Permian pole for the North American craton to date. It is statistically the same as only one other Permian pole, namely the Cap aux Meules pole (12) (Fig. 6) from the Iles de la Madeleine, about 100 km north of Prince Edward Island. Brisebois (1981) considered the Cap aux Meules red beds to be Early Permian and to be either correlative with the upper Pictou Group or to be stratigraphically above it. The paleomagnetic evidence indicates that they are correlative and that the Iles de la Madeleine have not been rotated counterclockwise during Permian time as Tanczyk (1988) proposed.

The upper Pictou pole falls directly on both the Irving and Irving (1982) and Gordon et al. (1984) APW paths. Although the age assignments along the latter path provide a better fit, the pole is reasonably consistent with both. It should be noted in passing that the Malpeque Bay sill pole (14) fits the Irving and Irving (1982) path better. Also the mean for the 11 Early Permian poles (3–13) is at 45°N, 121°E (4°), or halfway between poles 7 and 9. Again it falls on both paths but fits the Irving and Irving (1982) path better.

The distribution of Early Permian poles is curious. Those from eastern North America (11–13) are grouped distinctly to the southeast, at 41°N, 129°E (8°), of those (3–10) from the southwestern United States at 46°N, 117° (4°). There are several possible explanations. It could be simply chance, although the two groups are statistically different. It could reflect real age differences within the Early Permian, although this is incompatible with the apparent ages of several poles (e.g., 6 and 11). A more likely possibility is that the southwestern U.S. red beds were subjected to more prolonged hematization through the Late Permian or Early Triassic, so that their poles have been biased to younger ages on the APW path. Another hypothesis is that there has been about 9° of relative tectonic rotation between the two regions, presumably caused by the Laramide orogeny (Fig. 7). This hypothesis is consistent in magnitude with motions proposed for the southwestern United States by Hamilton (1981), Steiner (1986), and Bryan and Gordon (1986).

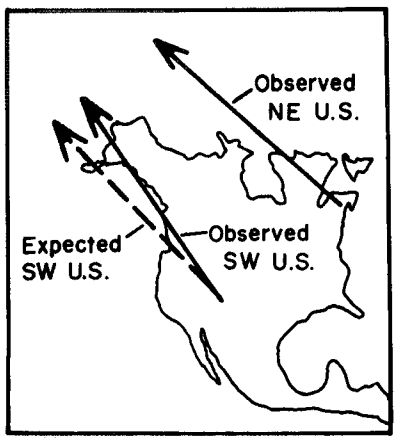


Figure 7. Apparent rotation from Early Permian paleomagnetic directions.

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